PennyPal — Pro Code Navigator & Live Defense Guide

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Purpose: Jump directly to any screen's code in 1 second during live viva defense and explain it like a senior developer.

Rule #1: Never Browse Folders with Your Mouse

Real developers do not click through folders on the left sidebar. They use keyboard shortcuts.

When a judge asks: "*Can you show the code for this screen or feature?*"

1. Press **Ctrl + P** in VS Code (Quick Open).
2. Type 3 to 5 letters of the file name (for example: **home_sc** or **budget_sc**).
3. Press **Enter**. The file opens instantly in 1 second.
4. If asked for a specific word or function, press **Ctrl + F** and type the keyword.

Doing this gives judges immediate confidence that you wrote the codebase and know every file by name.

Quick Jump Table: What to Type in Ctrl + P

Screen / Feature	Type in Ctrl + P	File Path	What to Point to on Screen
Main Dashboard & Header	`home_screen`	`lib/features/dashboard/presentation/home_screen.dart`	`_HomeHeader` (Avatar & Bell), `_BalanceHero`, `_PaceRing`
Monthly Budget & 1-Tap Copy	`budget_screen`	`lib/features/budgets/presentation/budget_screen.dart`	`_copyPreviousMonthBudget()` method
Add Expense & Gemini AI OCR	`transaction_form`	`lib/features/transactions/presentation/transaction_form_screen.dart`	`_scanReceiptWithGemini()` or `_pickReceiptImage()`
Offline Cache Engine	`main.dart`	`lib/main.dart` (lines 62–65)	`persistenceEnabled: true` & `CACHE_SIZE_UNLIMITED`
Savings Goals & Milestones	`savings_screen`	`lib/features/savings/presentation/savings_screen.dart`	`_GoalCard` & progress calculation
Split Bill	`cards_screen`	`lib/features/cards/presentation/cards_screen.dart`	Bill splitting calculation and WhatsApp share
Borrow & Lend (Debts)	`debt_tracker`	`lib/features/debts/presentation/debt_tracker_screen.dart`	`_DebtCard` (Lent vs Borrowed)
Subscriptions	`subscriptions_screen`	`lib/features/subscriptions/presentation/subscriptions_screen.dart`	Monthly recurring list & total cost calculation
Learning Hub & Lessons	`learning_screen`	`lib/features/learning/presentation/learning_screen.dart`	Lesson cards and categories
AI Assistant (Gemini Chat)	`assistant_screen`	`lib/features/assistant/presentation/assistant_screen.dart`	Chat message stream & Gemini prompt handler
Admin Panel	`admin_dashboard`	`lib/features/admin/presentation/admin_dashboard_screen.dart`	Admin statistics cards & charts
Navigation & Route Guards	`router.dart`	`lib/app/router.dart`	Role check: student vs admin redirection

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How Our Code is Structured (The 10-Second Architecture Pitch)

If judges ask: "*How did you structure your codebase?*", say:

> *"We structured PennyPal using **Feature-Driven Architecture**.*
> *Everything is inside `lib/features/`. Each feature has its own folder with:*
> • `presentation/` *for UI screens and widgets.*
> • `domain/` *for models and financial calculations.*
> • `data/` *for Firebase repositories and API calls."*

What to Say While Opening Code (Short & Natural)

Speak these simple sentences while your fingers type the shortcut:

- "*Sure, let me open that file directly.*" (press **Ctrl + P**)
- "*Here is our presentation widget.*"

- *"We separated UI from business logic."*
- *"State is handled through Riverpod providers."*
- *"Here is the calculation function."*

The 3 Most Likely Code Questions & How to Answer

Q1: "Show me how Offline Mode works in the code."

1. Press **Ctrl + P** ➔ type `main.dart` ➔ press **Enter**.
2. Scroll to lines 62–65:

```
FirebaseFirestore.instance.settings = const Settings(
  persistenceEnabled: true,
  cacheSizeBytes: Settings.CACHE_SIZE_UNLIMITED,
);
```

3. **What to say:**

> *"Right here in `main.dart`, we configure Firestore with `persistenceEnabled: true` and unlimited local cache. So every transaction writes to local device storage first, and syncs automatically when internet returns."*

Q2: "Show me where you call Gemini AI for receipts."

1. Press **Ctrl + P** ➔ type `transaction_form` ➔ press **Enter**.
2. Press **Ctrl + F** ➔ type `gemini` or `scan`.
3. **What to say:**

> *"Here in `transaction_form_screen.dart`, when the user picks a receipt photo, we send the image bytes to Gemini Vision AI. It extracts the merchant name, date, and amount, and populates the text controllers automatically."*

Q3: "Show me where the 1-Tap Quick Copy logic is."

1. Press **Ctrl + P** ➔ type `budget_screen` ➔ press **Enter**.
2. Press **Ctrl + F** ➔ type `copy`.
3. **What to say:**

> *"Here in `budget_screen.dart`, this function reads last month's budget documents from Firestore and creates duplicates for the current calendar month with one tap."*

Quick Tips for Live Confidence

- **Practice 3 times now:** Press **Ctrl + P** and jump between `home_screen.dart`, `budget_screen.dart`, and `main.dart`.
- **Stay relaxed:** If you misspell a file name, just press **Backspace** and retype. Never panic.
- **Point with your finger or mouse:** Once the file opens, highlight the class name or function with your mouse so the judge's eyes follow your lead.